

CISC 699 APPLIED PROJECT IN COMPUTER INFORMATION SCIENCES *SU26 - 3 Semester Hours*

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Delivery Mode: IP / HB / OS / OA
Online Meetings: Scheduled for All Formats - CISC 699 50 A & CISC 699 90 O

W 6:00 PM-8:00 PM	5/9/2026 - 8/14/2026	Online Virtual Classroom
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On Campus: Scheduled for Executive Visits - CISC 699 50 A

S 8:00 AM-1:00 PM	5/9/2026 - 5/9/2026	Harrisburg University Main Campus, 326 Market Street Building, 1412
S 8:00 AM-1:00 PM	7/25/2026 - 7/25/2026	Harrisburg University Main Campus, 326 Market Street Building, 1412

CATALOG ALIGNMENT

Catalog course description	This course allows the student to pursue an area of interest that is within the broad scope of Computer Information Sciences. A faculty member will supervise this study.
Course prerequisite	GRAD 695 - Research Methodology and Writing, or permission of instructor.
Program context	Within the 36-credit M.S. in Computer Information Sciences, CISC 699 is one of the program's two experiential-course options and follows GRAD 695 as the culminating applied-project experience.

COURSE OVERVIEW & ROLE IN THE DEGREE

CISC 699 functions as the culminating applied-project studio for the M.S. in Computer Information Sciences. Students synthesize prior coursework, concentration knowledge, and research-methods preparation into one substantial computational solution aimed at a meaningful technical, professional, organizational, or societal problem. The course is designed to reflect HU's highly engaged, collaborative, experiential, and applied graduate model while also meeting the expectations of a contemporary U.S. STEM master's capstone.

- Individual projects are the default. Team projects may be approved only when the scope genuinely requires collaboration and each student's responsibilities, technical contribution, and assessment evidence can be clearly separated.
- Projects may be faculty-sponsored, self-proposed, employer-sponsored, or community-oriented, provided they are academically appropriate, technically substantive, and feasible within one semester.
- A successful project produces both a functioning computational artifact and a professional-quality technical report that documents the problem, literature grounding, design choices, implementation, evaluation evidence, limitations, and next steps.
- Unacceptable project forms include tutorial replication, shallow product assembly without technical depth, or an unscoped software built with no evaluation plan.

COURSE PREREQUISITE

- GRAD 695 "Research Methodology & Writing", or permission of instructor

STUDENT LEARNING OUTCOMES (SLOs)

By the end of this course, students should be able to:

LO	What Students Demonstrate	HU Competency
1	Define an impactful applied problem, establish scope and success criteria, and justify why the project matters within computer information sciences.	Critical thinking / information literacy
2	Ground the project in relevant scholarly literature, technical documentation, standards, benchmarks, and prior solutions.	Critical thinking / information literacy

3	Design a computational solution with defensible decisions regarding architecture, algorithms, data, tools, interfaces, testing strategy, and constraints.	Critical thinking
4	Implement or substantially prototype a working software/system artifact appropriate to the approved project scope.	Critical thinking / entrepreneurship
5	Evaluate the solution using credible evidence such as experiments, test suites, benchmark results, comparative analysis, user feedback, or other discipline-appropriate validation.	Critical thinking / teamwork
6	Communicate the project clearly in a graduate-level technical report, formal presentation, and live or recorded demonstration.	Written & oral communication
7	Address ethics, privacy, security, accessibility, sustainability, and broader impact issues that are relevant to the project domain.	Ethical awareness / global awareness
8	Manage the project professionally through milestone planning, revision cycles, version control, documentation, and transparent reporting of limitations.	Teamwork / professionalism

INSTRUCTIONAL MODEL & SUPERVISION

- CISC 699 is organized as a milestone-driven capstone studio rather than a lecture-only course. Class meetings and faculty consultations are used for project review, technical troubleshooting, peer discussion, presentation practice, and approval of successive project milestones.
- The faculty member supervising the study is responsible for approving scope, monitoring progress, giving formative feedback, and determining whether the final package demonstrates appropriate master's-level rigor.
- Students are expected to maintain a current project plan, meet milestone deadlines, communicate emerging risks early, and revise their work based on feedback.
- A realistic workload expectation is at least 8-10 hours per week outside scheduled meetings for reading, design, implementation, testing, writing, and revision.

REQUIRED DELIVERABLES

Deliverable	Minimum expectation
Project charter / proposal	Problem statement, goals, scope boundaries, stakeholders or use context, success criteria, preliminary literature/related work, risks, timeline.
Design and review package	Requirements, architecture/design model, toolchain, data plan, test/evaluation plan, and updated bibliography.
Midpoint progress review	Working prototype or partial implementation, implementation status, issues log, revised scope or schedule, evidence of forward progress.
Final computational artifact	Working software/system artifact or approved equivalent, source code or repository link, setup instructions, dependencies, and technical documentation.
Final technical report	Graduate-level report documenting context, prior work, methodology, implementation, results, interpretation, limitations, ethics/impact, and next steps.
Final presentation and demo	Professional presentation and live or recorded demonstration with question-and-answer discussion.

ASSESSMENT AND GRADING

Graded component	Weight
Approved proposal and scope package	10%
Design review / literature / evaluation plan	15%
Midpoint progress review or demo	10%
Final computational artifact and technical package	30%
Final technical report	25%
Final presentation and defense/demo	10%

Official HU master's grading scale: A = 4.00, B = 3.00, C = 2.00, F = 0.00. A minimum cumulative GPA of 3.00 is required for graduation from a master's program.

RECOMMENDED EVALUATION DIMENSIONS

- Problem framing and significance: Problem is worthwhile, scoped realistically, and grounded in the field.
- Technical design rigor: Design choices are justified and aligned to project goals and constraints.
- Implementation quality: Artifact functions at an appropriate depth and shows engineering competence.
- Evaluation and evidence: Claims are supported with appropriate testing, experiments, or analytical evidence.
- Communication quality: Report and presentation are organized, clear, professional, and technically accurate.
- Ethics and broader impact: Relevant privacy, security, accessibility, and societal implications are addressed.
- Professional process: Student manages milestones well, responds to feedback, and documents work reproducibly.

SUGGESTED 14-WEEK SCHEDULE

W	Focus / milestone
1	Orientation; review of course expectations; project selection discussion; supervisor assignment
	Problem framing; project charter draft; scope negotiation; initial bibliography and feasibility review.
2	Proposal approval checkpoint; success criteria finalized; initial project plan submitted.
3	Literature and related-work synthesis; requirements, use cases, data sources, or domain constraints
4	Architecture/design review; methods, environment, and testing/evaluation plan approved.
5	Implementation sprint I; repository setup; documentation baseline; risk and issue log update.
6	Implementation sprint II; data/toolchain validation; early evidence collection.
7	Midpoint demo or technical progress review; scope adjustment if needed.
8	Implementation sprint III; deeper testing, benchmarking, or comparative analysis.
9	Results development; limitations, error analysis, and broader-impact issues drafted.
10	Artifact hardening; README, setup instructions, and reproducibility materials completed.
11	Full draft of the final technical report due for feedback.
12	Revision cycle; presentation deck and demo script developed; final testing completed.
13	Formal presentation rehearsal; final package check against submission requirements.
14	Final report, computational artifact, and presentation/demo submission; course closeout and reflection.

STANDARDS FOR THE FINAL SUBMISSION

- A functioning computational artifact appropriate to the approved project type, together with a repository link or submission package as directed by the instructor.
- A final technical report, typically 15-25 pages excluding appendices, written in clear graduate-level English with consistent citations, figures, tables, and technical formatting.
- A README containing setup instructions, dependencies, execution steps, input/output expectations, and known limitations.
- Versioned source files, scripts, notebooks, configuration files, test assets, or other materials needed to understand or reproduce the project at a reasonable level.
- Testing or validation evidence such as unit/integration tests, benchmark tables, ablation or comparison results, screenshots, logs, user feedback, or analytical notes, as appropriate.
- A brief data statement describing source provenance, licensing, privacy constraints, and any synthetic, public, proprietary, or anonymized data decisions.
- An AI-use disclosure appendix whenever generative AI or similar systems were used for ideation, coding assistance, debugging, drafting, or analysis support.

RECOMMENDED FINAL REPORT ORGANIZATION

- Title and abstract
- Problem statement and significance
- Related work / background
- System or solution design
- Implementation details
- Evaluation methodology

- Results and analysis
- Ethics, security, privacy, accessibility, and broader impact
- Limitations and future work
- References and appendices

COURSE POLICIES

- Attendance is critical to academic success, and students are expected to demonstrate regular academic engagement in the course.
- In online sections, participation means substantive course activity such as submitting assignments, discussion posts, quizzes, tests, chats, or team sessions; logging in and viewing materials alone does not count as participation.
- Because this is a milestone-driven capstone, missed reviews or silent non-participation can seriously impair project success. If a student is absent for an extended period and fails to complete required participatory work, the instructor may follow the university's academic-withdrawal process.

Add/drop, withdrawal, and incomplete grades

- The university add/drop period begins on the first day of the semester or subterm and ends after six days of classes have occurred, including Saturday. Official withdrawal deadlines are governed by the Academic Calendar.
- After the add/drop period, a course withdrawal receives a W until the published deadline; after that point, a WF may be recorded according to university policy.
- An Incomplete (I) may be granted only at the instructor's discretion when documented circumstances beyond the student's control prevent completion and there is a reasonable likelihood that the work can be completed successfully.

Academic integrity and responsible use of AI

- HU expects students to submit work that is genuinely their own and to refrain from all forms of lying, cheating, and plagiarism.
- AI tools may be used only in ways permitted by the instructor. Any substantial AI assistance must be disclosed transparently. Students remain fully responsible for the correctness, originality, legality, and appropriateness of all submitted work.
- Students may not submit AI-generated text, code, analysis, or figures as if these were wholly their own unaided work, and they may not use public AI systems with confidential, proprietary, personal, FERPA-protected, HIPAA-regulated, or otherwise restricted information unless formally authorized

Accessibility and student support

- Students seeking accommodation should work through Accessibility Support Services. Questions regarding documentation and services may be directed to ADA@harrisburgu.edu.
- Students needing academic coaching or broader student-support guidance may work with Graduate Student Success / the Office of Student Success.
- Students are encouraged to contact the instructor early if challenges arise that may affect project progress.

Professional communication and tools

- Official communication should use HU email, Canvas, MyHU, and other university-approved systems.
- Students are expected to maintain professional versioning and documentation practices. Git-based source control, issue tracking, and clearly labeled submission folders are strongly recommended.
- Projects that involve human participants, protected data, external sponsors, or regulated/security-sensitive environments must satisfy all applicable approvals and restrictions before implementation or data collection begins.

International student note

- Students on F-1 status remain responsible for maintaining eligibility requirements and should consult the International Student Office regarding attendance, enrollment, and CPT-related questions tied to experiential study.

KEY HU SUPPORT RESOURCES

Office / service	Primary contact	Use case
Graduate Student Success	GradHelp Student Portal / 717-901-5102	academic coaching, student support, navigation of university services
Accessibility Support Services	ADA@harrisburgu.edu	accommodations, disability documentation, access support
Library	library@harrisburgu.libanswers.com	research guidance, databases, literature search support

Information Technology	ithelp.harrisburgu.edu / 717-901-5177	Canvas, MyHU, access, software, technical issues
Career Services & Experiential Learning	CareerServices@HarrisburgU.edu	professional development, internships, employer-facing support
Records and Registration	reghelp.harrisburgu.edu / 717-901-5136	registration, withdrawal, academic record questions
CARES / student wellness	CARES@harrisburgu.edu	wellness and support referral

RECOMMENDED REFERENCES AND RESOURCES

- Research-methods and graduate writing resources appropriate to GRAD 695 and the student's project area.
- Current peer-reviewed literature, standards documents, benchmark papers, frameworks, SDK/API documentation, and technical manuals relevant to the approved topic.
- Department guidance on code quality, software testing, data governance, cybersecurity, responsible AI use, and scholarly citation.
- Selected final-project outputs may be archived or adapted for the student's ePortfolio or other evidence-based graduation materials when appropriate.

PUBLISHING CHECKLIST BEFORE OFFICIAL USE

- Insert instructor name, HU email, office location, office hours, semester, section number, and exact meeting information.
- Confirm whether the section is in-person, hybrid, synchronous online, or asynchronous online, and align attendance language accordingly.
- Confirm any department-required report template, repository submission standard, or presentation format.
- Add any current term-specific student handbook statements, Title IX language, religious observance guidance, military-service language, or other boilerplate required by HU.

Honor Code - We as members of Harrisburg University community pledge not to cheat, plagiarize, steal, or lie in matters related to academic work. As a Community of Learners, we honor and uphold the *HU Honor Code*.

Note: Any student involved in plagiarism will automatically be given a grade of zero and the student will be reported to the HU administration.